



9438 - Certum: Multiband Webb Images in the Inner Galaxy for the Roman Galactic Plane Survey

Cycle: 4, Proposal Category: DD

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Eddie Schlafly (PI)	Space Telescope Science Institute
Dr. Joshua Peek (CoI)	Space Telescope Science Institute
Dr. Adam Smercina (CoI)	Space Telescope Science Institute
Dr. Robert A. Benjamin (CoI)	University of Wisconsin - Whitewater
Dr. Erik Tollerud (CoI)	Space Telescope Science Institute
Andrew Saydjari (CoI)	Princeton University
Dr. Kristen B W McQuinn (CoI)	Space Telescope Science Institute
Catherine Zucker (CoI)	Smithsonian Institution Astrophysical Observatory

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
imaging				
	1		NIRCam Imaging	(1) G028.320+01.243
	2		NIRCam Imaging	(2) G033.007+01.150
	3		NIRCam Imaging	(3) G040.954+02.473
	4		NIRCam Imaging	(4) G054.093+01.748
	5		NIRCam Imaging	(5) G007.470+00.050
	6		NIRCam Imaging	(6) crowded_l3
	7		NIRCam Imaging	(7) crowded_l20

ABSTRACT

The Roman Galactic Plane Survey (RGPS) will observe ~ 700 square degrees of the inner Galaxy, unveiling much of the far side of the Milky Way's disk and measuring fluxes for tens of billions of stars. The success of the RGPS depends on understanding the connection between the photometry Roman measures and the Galaxy's stars, gas, and dust, in the extremely crowded and reddened environments typical of the inner galaxy. This is a major challenge: while we expect that the far side of the Galaxy will look broadly like the near side, it is (1) largely unobserved today at Roman's sensitivity, (2) heavily reddened by dust, and (3) very crowded. This proposal, Certum, will obtain Webb imaging of seven RGPS fields over a range of environments, with the much better spectral coverage, depth, and resolution made possible by Webb. We will use the Certum observations to optimize the RGPS design, quantify the ability of the RGPS to measure the properties of the Galaxy's stars and dust, and to determine the effect of crowding on Roman observations. Additionally, Certum will specifically target fields featuring gas-bearing regions on the far side of the Galaxy in order to map the structure of the Milky Way and to test the RGPS's mapping potential.

OBSERVING DESCRIPTION

We propose observations of 7 inner galaxy fields in 14 filters with NIRCcam (F070W, F090W, F115W, F140M, F150W, F182M, F210M, F277W, F300M, F335M, F360M, F410M, F444W, F480M). Each field is observed with 10 groups of the SHALLOW4 readout pattern, using a 4-point INTRAMODULEBOX dither pattern. The observations will be used to provide reference images and catalogs for planning the Roman Galactic Plane Survey, and to map the three-dimensional distribution of dust on the far side of the galaxy in the outer Scutum-Centaurus arm.

Proposal 9438 - Targets - Certum: Multiband Webb Images in the Inner Galaxy for the Roman Galactic Plane Survey

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	G028.320+01.243	RA: 18 38 34.8000 (279.6450000d) Dec: -03 32 6.00 (-3.53500d) Equinox: J2000	Parallax: 0" Epoch of Position: 2000	
	Comments: From Amentrout+2017 Category=Stellar Cluster Description=[Young star clusters] Extended=YES				
	(2)	G033.007+01.150	RA: 18 47 28.8000 (281.8700000d) Dec: +00 35 27.60 (.59100d) Equinox: J2000	Epoch of Position: 2000	
	Comments: From Amentrout+2017 Category=Stellar Cluster Description=[Young star clusters] Extended=YES				
	(3)	G040.954+02.473	RA: 18 57 16.0800 (284.3170000d) Dec: +08 15 57.60 (8.26600d) Equinox: J2000	Epoch of Position: 2000	
	Comments: From Amentrout+2017 Category=Stellar Cluster Description=[Young star clusters] Extended=YES				
	(4)	G054.093+01.748	RA: 19 24 58.5600 (291.2440000d) Dec: +19 34 30.00 (19.57500d) Equinox: J2000	Epoch of Position: 2000	
	Comments: From Amentrout+2017 Category=Stellar Cluster Description=[Young associations] Extended=YES				
	(5)	G007.470+00.050	RA: 18 02 11.0760 (270.5461500d) Dec: -22 28 20.57 (-22.47238d) Equinox: J2000	Parallax: 0" Epoch of Position: 2000	
	Comments: From Sanna+2017. Category=Stellar Cluster Description=[Young star clusters] Extended=YES				
	(6)	crowded_l3	RA: 17 52 35.7600 (268.1490000d) Dec: -26 21 50.40 (-26.36400d) Equinox: J2000		
	Comments: "blank" field at Galactic coordinates (l, b) = (3, 0). Category=Unidentified Description=[Blank field] Extended=YES				
	(7)	crowded_l20	RA: 18 27 31.6800 (276.8820000d) Dec: -11 29 20.40 (-11.48900d) Equinox: J2000	Epoch of Position: 2000	
	Comments: "Blank field" at Galactic coordinates (l, b) = (20, 0). Category=Unidentified Description=[Blank field] Extended=YES				

Proposal 9438 - Observation 1 - Certum: Multiband Webb Images in the Inner Galaxy for the Roman Galactic Plane Survey

Observation	Proposal 9438, Observation 1										Tue Aug 19 20:00:29 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	G028.320+01.243	RA: 18 38 34.8000 (279.6450000d)			Parallax: 0"					
	Dec: -03 32 6.00 (-3.53500d)			Epoch of Position: 2000							
	Equinox: J2000										
	Comments: From Amentrout+2017 Category=Stellar Cluster Description=[Young star clusters] Extended=YES										
Template	Module			Subarray			Target Placement				
	ALL			FULL			Module A (A3 corner)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F210M	F410M	SHALLOW4	5	1	4	4	1030.73	250756.6	
	2	F140M	F430M	SHALLOW4	5	1	4	4	1030.73	250756.8	
	3	F182M	F480M	SHALLOW4	5	1	4	4	1030.73	250756.5	
	4	F150W	F360M	SHALLOW4	5	1	4	4	1030.73	250756.4	
	5	F115W	F335M	SHALLOW4	5	1	4	4	1030.73	250756.3	
	6	F090W	F300M	SHALLOW4	5	1	4	4	1030.73	250756.2	
	7	F070W	F277W	SHALLOW4	5	1	4	4	1030.73	250756.1	
Special Requirements	Before Date 30-NOV-2025:00:00:00 Aperture PA Range 28.86565232 to 98.86565232 Degrees (V3 29.0 to 99.0) Aperture PA Range 108.86565232 to 109.86565232 Degrees (V3 109.0 to 110.0) Fiducial Point Override NRCAS_FULL										

Proposal 9438 - Observation 2 - Certum: Multiband Webb Images in the Inner Galaxy for the Roman Galactic Plane Survey

Observation	Proposal 9438, Observation 2										Tue Aug 19 20:00:29 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	G033.007+01.150	RA: 18 47 28.8000 (281.8700000d)			Epoch of Position: 2000					
			Dec: +00 35 27.60 (.59100d)								
			Equinox: J2000								
	Comments: From Amentrout+2017										
	Category=Stellar Cluster										
	Description=[Young star clusters]										
	Extended=YES										
Template	Module			Subarray			Target Placement				
	ALL			FULL			Module A (A3 corner)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F210M	F410M	SHALLOW4	5	1	4	4	1030.73	250756.6	
	2	F140M	F430M	SHALLOW4	5	1	4	4	1030.73	250756.8	
	3	F182M	F480M	SHALLOW4	5	1	4	4	1030.73	250756.5	
	4	F150W	F360M	SHALLOW4	5	1	4	4	1030.73	250756.4	
	5	F115W	F335M	SHALLOW4	5	1	4	4	1030.73	250756.3	
	6	F090W	F300M	SHALLOW4	5	1	4	4	1030.73	250756.2	
	7	F070W	F277W	SHALLOW4	5	1	4	4	1030.73	250756.1	
Special Requirements	Before Date 30-NOV-2025:00:00:00										
	Fiducial Point Override NRCAS_FULL										

Proposal 9438 - Observation 3 - Certum: Multiband Webb Images in the Inner Galaxy for the Roman Galactic Plane Survey

Observation	Proposal 9438, Observation 3										
	Diagnostic Status: Warning Observing Template: NIRCam Imaging										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous			
	(3)	G040.954+02.473	RA: 18 57 16.0800 (284.3170000d) Dec: +08 15 57.60 (8.26600d) Equinox: J2000			Epoch of Position: 2000					
Template	Comments: From Amentrout+2017 Category=Stellar Cluster Description=[Young star clusters] Extended=YES										
	Module		Subarray				Target Placement				
	ALL		FULL				Module A (A3 corner)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F210M	F410M	SHALLOW4	5	1	4	4	1030.73	250756.6	
	2	F140M	F430M	SHALLOW4	5	1	4	4	1030.73	250756.8	
	3	F182M	F480M	SHALLOW4	5	1	4	4	1030.73	250756.5	
	4	F150W	F360M	SHALLOW4	5	1	4	4	1030.73	250756.4	
	5	F115W	F335M	SHALLOW4	5	1	4	4	1030.73	250756.3	
	6	F090W	F300M	SHALLOW4	5	1	4	4	1030.73	250756.2	
	7	F070W	F277W	SHALLOW4	5	1	4	4	1030.73	250756.1	
Special Requirements	Before Date 30-NOV-2025:00:00:00 Fiducial Point Override NRCAS_FULL										

Proposal 9438 - Observation 4 - Certum: Multiband Webb Images in the Inner Galaxy for the Roman Galactic Plane Survey

Observation	Proposal 9438, Observation 4										Tue Aug 19 20:00:29 GMT 2025
	Diagnostic Status: Warning Observing Template: NIRCam Imaging										
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	G054.093+01.748	RA: 19 24 58.5600 (291.2440000d) Dec: +19 34 30.00 (19.57500d) Equinox: J2000			Epoch of Position: 2000					
Template	Comments: From Amentrout+2017 Category=Stellar Cluster Description=[Young associations] Extended=YES										
	Module	Subarray				Target Placement					
	ALL	FULL				Module A (A3 corner)					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F210M	F410M	SHALLOW4	5	1	4	4	1030.73	250756.6	
	2	F140M	F430M	SHALLOW4	5	1	4	4	1030.73	250756.8	
	3	F182M	F480M	SHALLOW4	5	1	4	4	1030.73	250756.5	
	4	F150W	F360M	SHALLOW4	5	1	4	4	1030.73	250756.4	
	5	F115W	F335M	SHALLOW4	5	1	4	4	1030.73	250756.3	
	6	F090W	F300M	SHALLOW4	5	1	4	4	1030.73	250756.2	
	7	F070W	F277W	SHALLOW4	5	1	4	4	1030.73	250756.1	
Special Requirements	Before Date 30-NOV-2025:00:00:00 Aperture PA Range 359.86565232 to 133.86565232 Degrees (V3 0.0 to 134.0) Fiducial Point Override NRCAS_FULLL										

Proposal 9438 - Observation 5 - Certum: Multiband Webb Images in the Inner Galaxy for the Roman Galactic Plane Survey

Observation	Proposal 9438, Observation 5										Tue Aug 19 20:00:29 GMT 2025
	Diagnostic Status: Warning Observing Template: NIRCam Imaging										
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	G007.470+00.050	RA: 18 02 11.0760 (270.5461500d) Dec: -22 28 20.57 (-22.47238d) Equinox: J2000			Parallax: 0" Epoch of Position: 2000					
Template	Comments: From Sanna+2017. Category=Stellar Cluster Description=[Young star clusters] Extended=YES										
	Module		Subarray				Target Placement				
Dithers	ALL		FULL				Module A (A3 corner)				
	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
Spectral Elements	1	INTRAMODULEBOX		4		STANDARD				1	
	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F210M	F410M	SHALLOW4	5	1	4	4	1030.73	250756.6	
	2	F140M	F430M	SHALLOW4	5	1	4	4	1030.73	250756.8	
	3	F182M	F480M	SHALLOW4	5	1	4	4	1030.73	250756.5	
	4	F150W	F360M	SHALLOW4	5	1	4	4	1030.73	250756.4	
	5	F115W	F335M	SHALLOW4	5	1	4	4	1030.73	250756.3	
	6	F090W	F300M	SHALLOW4	5	1	4	4	1030.73	250756.2	
	7	F070W	F277W	SHALLOW4	5	1	4	4	1030.73	250756.1	
Special Requirements	Before Date 30-NOV-2025:00:00:00 Aperture PA Range 88.86565232 to 194.86565232 Degrees (V3 89.0 to 195.0) Fiducial Point Override NRCAS_FULLL										

Proposal 9438 - Observation 6 - Certum: Multiband Webb Images in the Inner Galaxy for the Roman Galactic Plane Survey

Observation	Proposal 9438, Observation 6									Tue Aug 19 20:00:29 GMT 2025	
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous			
	(6)	crowded_l3	RA: 17 52 35.7600 (268.1490000d) Dec: -26 21 50.40 (-26.36400d) Equinox: J2000								
	Comments: "blank" field at Galactic coordinates (l, b) = (3, 0). Category=Unidentified Description=[Blank field] Extended=YES										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module A (A3 corner)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F210M	F410M	SHALLOW4	5	1	4	4	1030.73	250756.6	
	2	F140M	F430M	SHALLOW4	5	1	4	4	1030.73	250756.8	
	3	F182M	F480M	SHALLOW4	5	1	4	4	1030.73	250756.5	
	4	F150W	F360M	SHALLOW4	5	1	4	4	1030.73	250756.4	
	5	F115W	F335M	SHALLOW4	5	1	4	4	1030.73	250756.3	
	6	F090W	F300M	SHALLOW4	5	1	4	4	1030.73	250756.2	
	7	F070W	F277W	SHALLOW4	5	1	4	4	1030.73	250756.1	
Special Requirements	Before Date 30-NOV-2025:00:00:00 Fiducial Point Override NRCAS_FULL										

Proposal 9438 - Observation 7 - Certum: Multiband Webb Images in the Inner Galaxy for the Roman Galactic Plane Survey

Observation	Proposal 9438, Observation 7										Tue Aug 19 20:00:29 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCcam Imaging										
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(7)	crowded_l20	RA: 18 27 31.6800 (276.8820000d)			Epoch of Position: 2000					
			Dec: -11 29 20.40 (-11.48900d)								
			Equinox: J2000								
		Comments: "Blank field" at Galactic coordinates (l, b) = (20, 0). Category=Unidentified Description=[Blank field] Extended=YES									
Template	Module			Subarray			Target Placement				
	ALL			FULL			Module A (A3 corner)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F210M	F410M	SHALLOW4	5	1	4	4	1030.73	250756.6	
	2	F140M	F430M	SHALLOW4	5	1	4	4	1030.73	250756.8	
	3	F182M	F480M	SHALLOW4	5	1	4	4	1030.73	250756.5	
	4	F150W	F360M	SHALLOW4	5	1	4	4	1030.73	250756.4	
	5	F115W	F335M	SHALLOW4	5	1	4	4	1030.73	250756.3	
	6	F090W	F300M	SHALLOW4	5	1	4	4	1030.73	250756.2	
	7	F070W	F277W	SHALLOW4	5	1	4	4	1030.73	250756.1	
Special Requirements	Before Date 30-NOV-2025:00:00:00 Fiducial Point Override NRCAS_FULL										